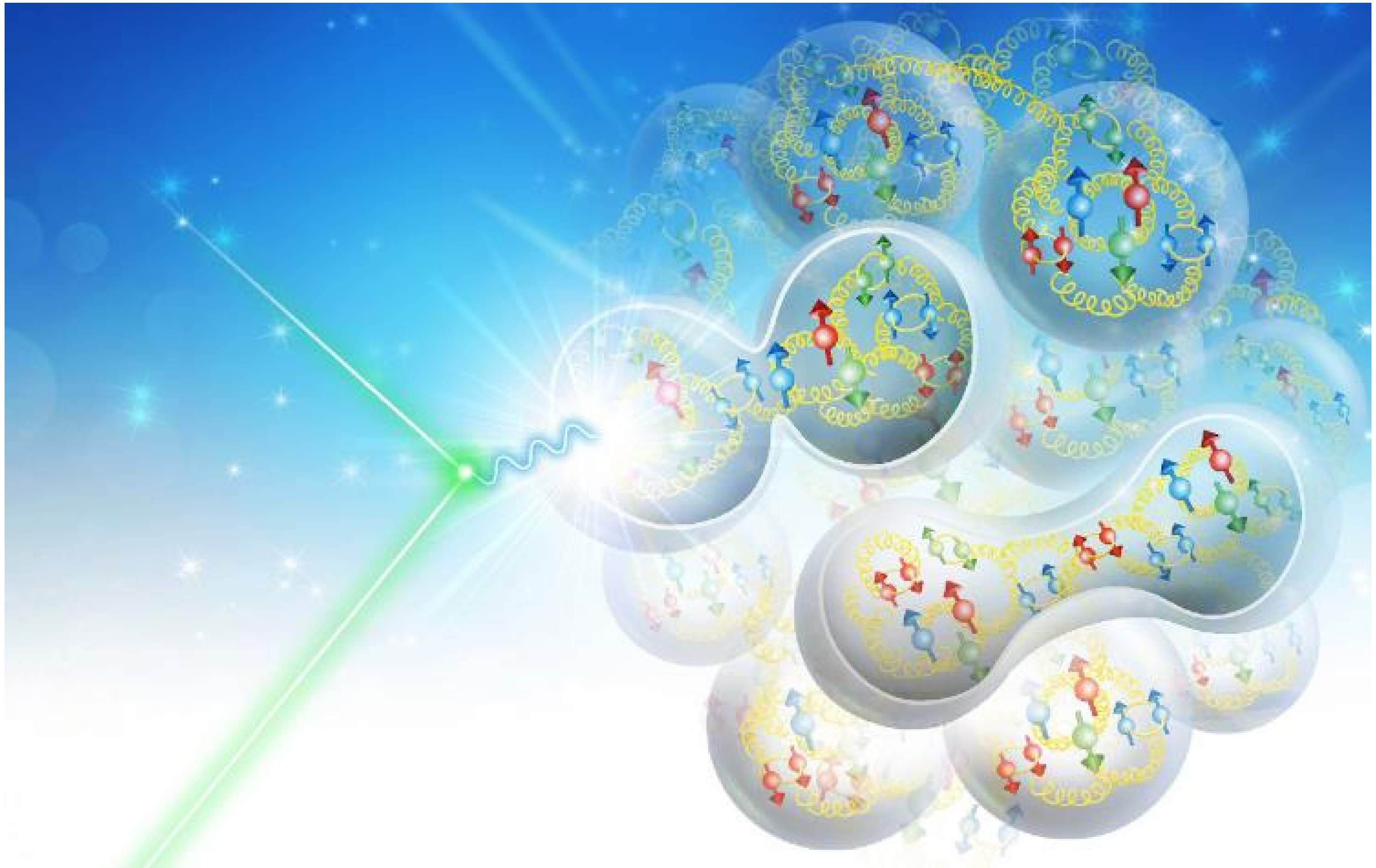
Interaction Region Magnets for Future Electron-Ion Collider at Jefferson Lab



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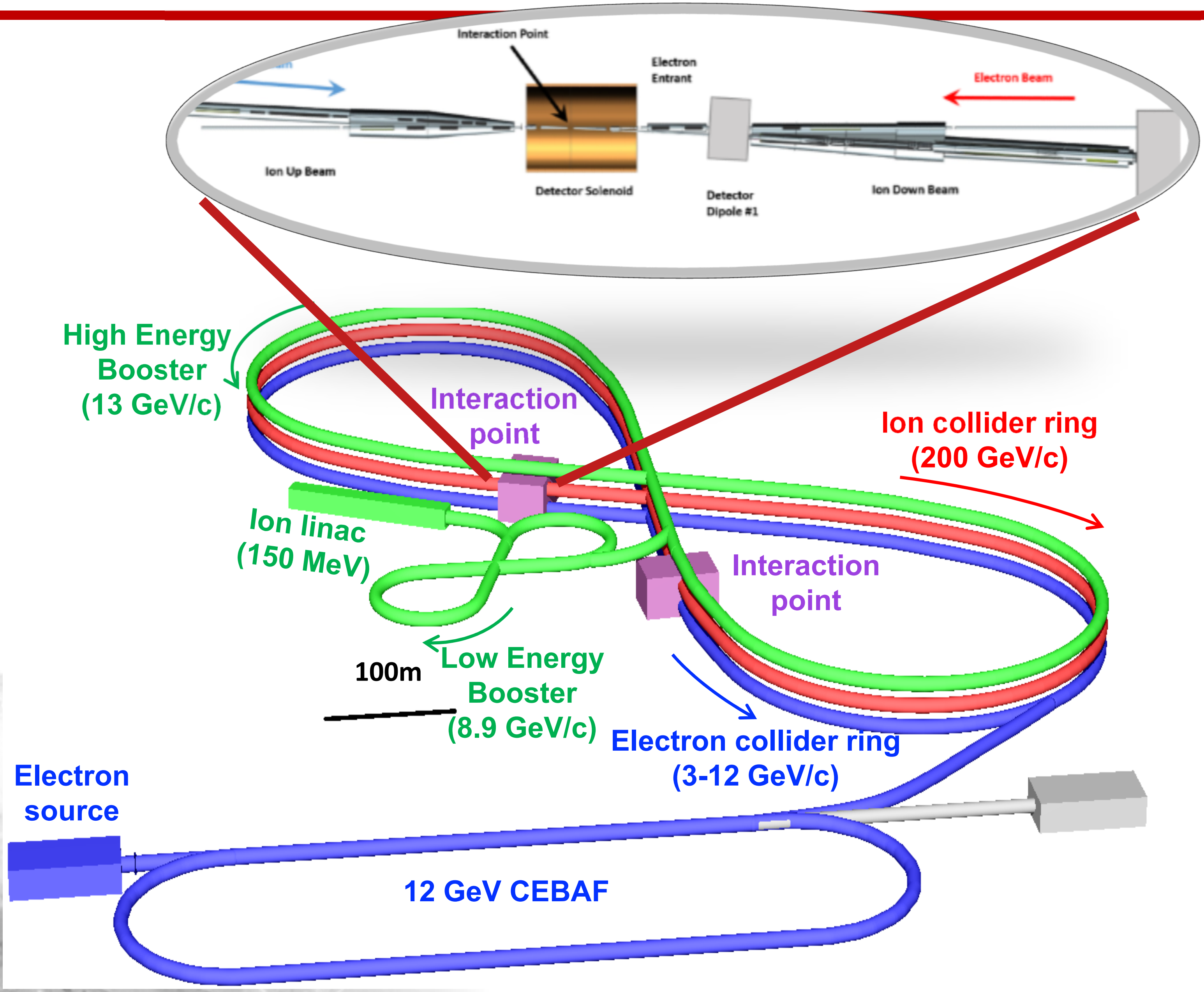
JLEIC Overview



- **Electron complex**
  - CEBAF as a full energy injector
  - Electron collider ring: 3-12 GeV/c
- **Ion complex**
  - Ion source, SRF linac: 150 MeV for protons
  - Low Energy Booster: 8.9 GeV/c
  - High Energy Booster: 13 GeV/c
  - Ion collider ring: 200 GeV/c
- **Up to two detectors** at minimum background locations
- **20-100 GeV CM, Upgradable to 140 GeV CM**

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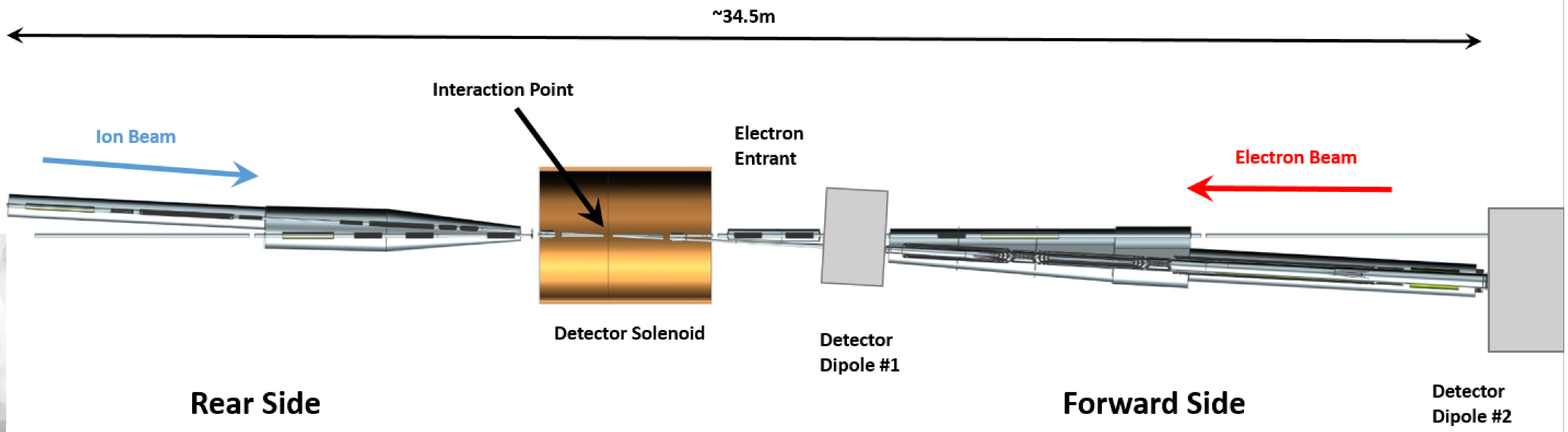
Interaction Region

- The electron and ion rings intersect at the interaction point (IP) and the region around IP is called interaction region (IR).
- Crossing angle is 50 mrad
- The IR contains a full acceptance detector built around a detector solenoid.
- Forward Side ion magnets have apertures which support ±10 mrad angular acceptance



This presentation does not cover the detector solenoid or the detector dipoles.

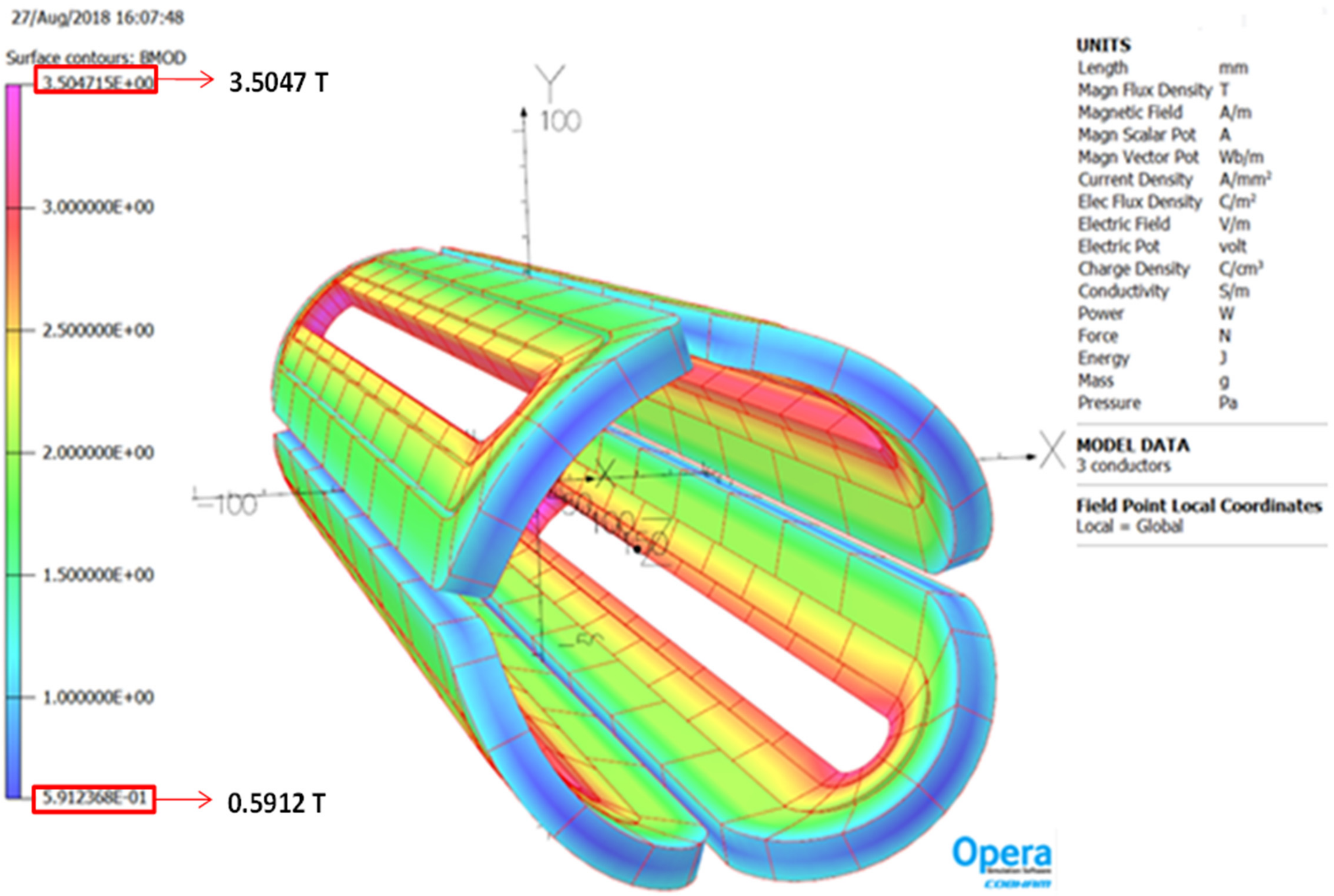
IR Magnet Specifications and Design Parameters

- All final focus quadrupole magnets use NbTi conductor
- Operating temperature between 4.5 K and 4.7 K
- All IR magnets are designed as cold bore, to lower the peak field in the coils

		Specifications									Design			
Element Name	Type	Length [m]	Good Field Radius [cm]	Aperture Inner Radius [cm]	Outer Radius [cm]	Dipole field [T]		Quadrupole field [T/m]		Solenoid [T]	Coil Inner Radius (cm)	Coil Outer Radius (cm)	Coil Width in Radial Direction (mm)	Peak Field in the coil (T)
Ion Rear Side Elements						Bx	By	Normal	Skew					
iASUS	SOLENOID	1.6	3.0	4.0	12.0	0	0	0	0	2	6	6.7	7	2.0
iQUS3S	QUADRUPOLE	0.5	3.0	4.0	12.0	0	0	0	3.38	0	4.5	4.7	2	0.3
iQUS2	QUADRUPOLE	2.1	3.0	4.0	12.0	0	0	94.07	0	0	4.5	7.8	33	5.7
iQUS2S	QUADRUPOLE	0.5	2.0	3.0	10.0	0	0	0	-9.26	0	3.5	4	5	0.6
iQUS1b	QUADRUPOLE	1.45	2.0	3.0	10.0	0	0	-97.88	0	0	3.45	4.95	15	5.1
iQUS1S	QUADRUPOLE	0.5	2.0	3.0	10.0	0	0	0	16.42	0	3.5	4.4	9	0.9
iQUS1a	QUADRUPOLE	1.45	2.0	3.0	10.0	0	0	-97.88	-3.08	0	3.45	5.75	23	5.1
iCUS1	KICKER	0.3	2.0	3.0	10.0	-3.90	0.076	0	0	0	3.45	5.25	18	6.3
iCUS2	KICKER	0.3	2.0	3.0	10.0	4.50	-0.019	0	0	0				
Ion Forward Side Elements														
iQDS1a	QUADRUPOLE	2.25	4.0	9.2	23.1	0.0	0	-37.23	-1.23	0	13.0	17.1	41	6.4
iQDS1S	QUADRUPOLE	0.5	4.0	9.9	24.8	0.0	0	0	14.85	0	13.0	14.2	12	3.9
iQDS1b	QUADRUPOLE	2.25	4.0	12.3	31.0	0.0	0	-37.23	0	0	13.0	16.3	33	6.4
iQDS2S	QUADRUPOLE	0.5	4.0	13.0	32.7	0.0	0	0	-7.83	0	13.6	14.5	9	2.3
iQDS2	QUADRUPOLE	4.5	4.0	17.7	44.4	0.0	0	25.96	0	0	18.2	21.5	33	7.0
iQDS3S	QUADRUPOLE	0.5	4.0	18.4	46.2	0.0	0	0	0.63	0	20.0	20.2	2	0.4
iASDS	SOLENOID	1.2	4.0	19.8	49.7	0.0	0	0	0	4	22.5	24.0	15	4.0
Electron Rear Side Elements														
eASDS	SOLENOID	1.2	2.2	4.5	11.0	0	0	0	0	-4	6.5	8.0	15	4.0
eQDS3	QUADRUPOLE	0.6	2.4	4.5	10.0	0	0	-18.72	-2.71	0	4.95	6.5	15.5	3.6
eQDS2	QUADRUPOLE	0.6	2.8	4.5	8.5	0	0	36.22	5.25	0				
eQDS1	QUADRUPOLE	0.6	1.7	4.5	8.0	0	0	-33.75	-4.89	0				
Electron Forward Side Elements														
eQUS1	QUADRUPOLE	0.6	2.0	4.5	10.0	0.0	0.00	-36.94	8.10	0	4.95	6.5	15.5	3.6
eQUS2	QUADRUPOLE	0.6	3.2	4.5	11.0	0.0	0.00	33.66	-7.38	0				
eQUS3	QUADRUPOLE	0.6	1.5	4.5	11.0	0.0	0	-20.80	4.56	0				
eASUS	SOLENOID	1.8	2.2	4.5	11.0	0.0	0	0	0	-4	6.5	8.0	15	4.0

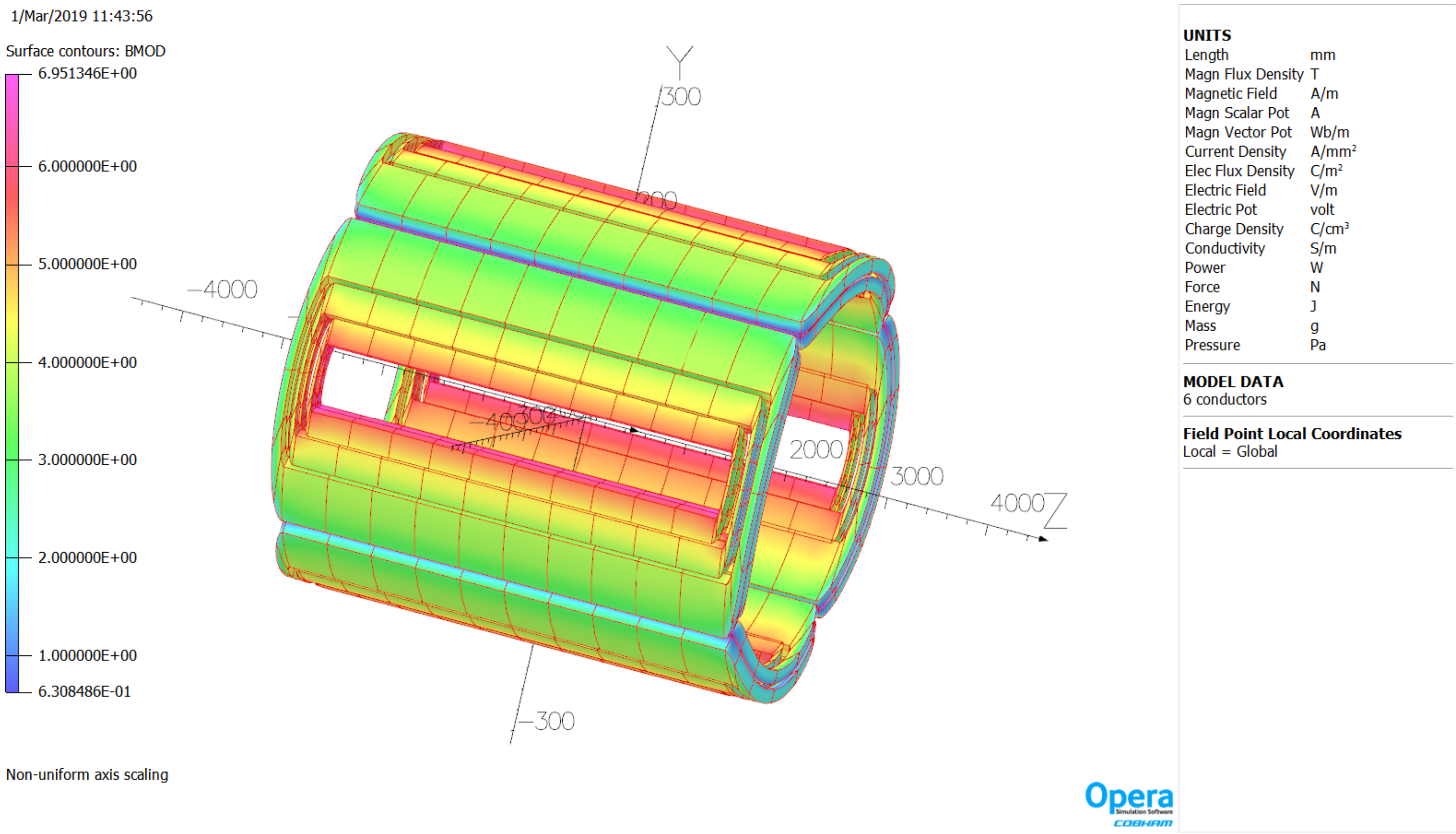
Electron Quadrupoles

- The electron beam line has 6 main quadrupoles
- Field gradient varies but all the magnets are designed for 45 T/m gradient
- A common coil design is used for these 6 quadrupoles
- The peak field in the coil is 3.6 T
- The coils will be keystone Rutherford cable
- Further optimization: multipole effects, field in the coil, conductor selection



Ion Quadrupole

- The ion beam line has 6 main quadrupoles; 3 upstream, 3 downstream
- The upstream quadrupoles are less demanding due to their smaller apertures
- The downstream quadrupoles have larger bores and are the most challenging
  - iQDS1a has a gradient of 37.23 T/m and beam aperture radius of 9.2 cm
  - iQDS1b has a gradient of 37.23 T/m and beam aperture radius of 12.3 cm
  - iQDS2 has a gradient of 25.96 T/m and the largest beam aperture radius at 17.7 cm
- The peak field in the ion quadrupole coils is 6.4-7.0 T
- The coils will be keystone Rutherford cable



This coil design is optimized to the first order. Further optimization, is likely to reduce the peak field in the coils.

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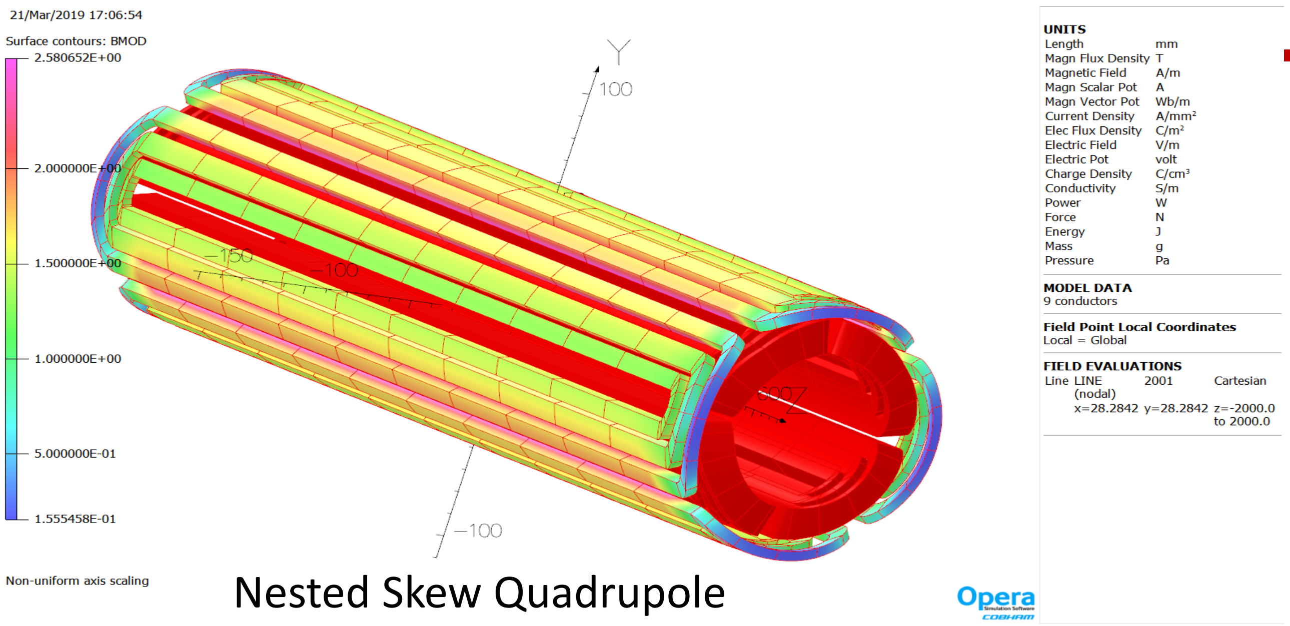
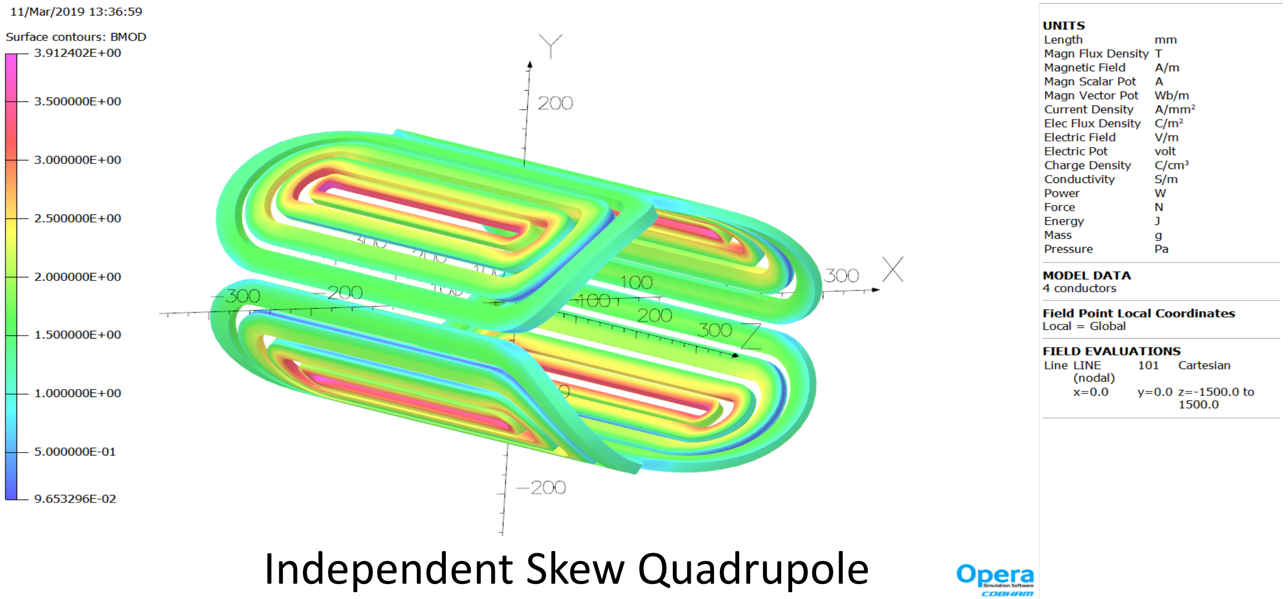
Skew Quadrupole

- Electron beam line skew quadrupoles are nested around the main quadrupole
- The first ion upstream and downstream skew quadrupoles are also nested coils
- The other ion beam line skew quadrupoles will be independent due to the required gradient
- The coil field in all the skew quadrupoles is relatively low to moderate for NbTi
- The peak field in the coil for an independent skew quadrupole (iQDS1S) is 3.9 T
- Skew quadrupole conductor will be standard MRI rectangular or round conductor



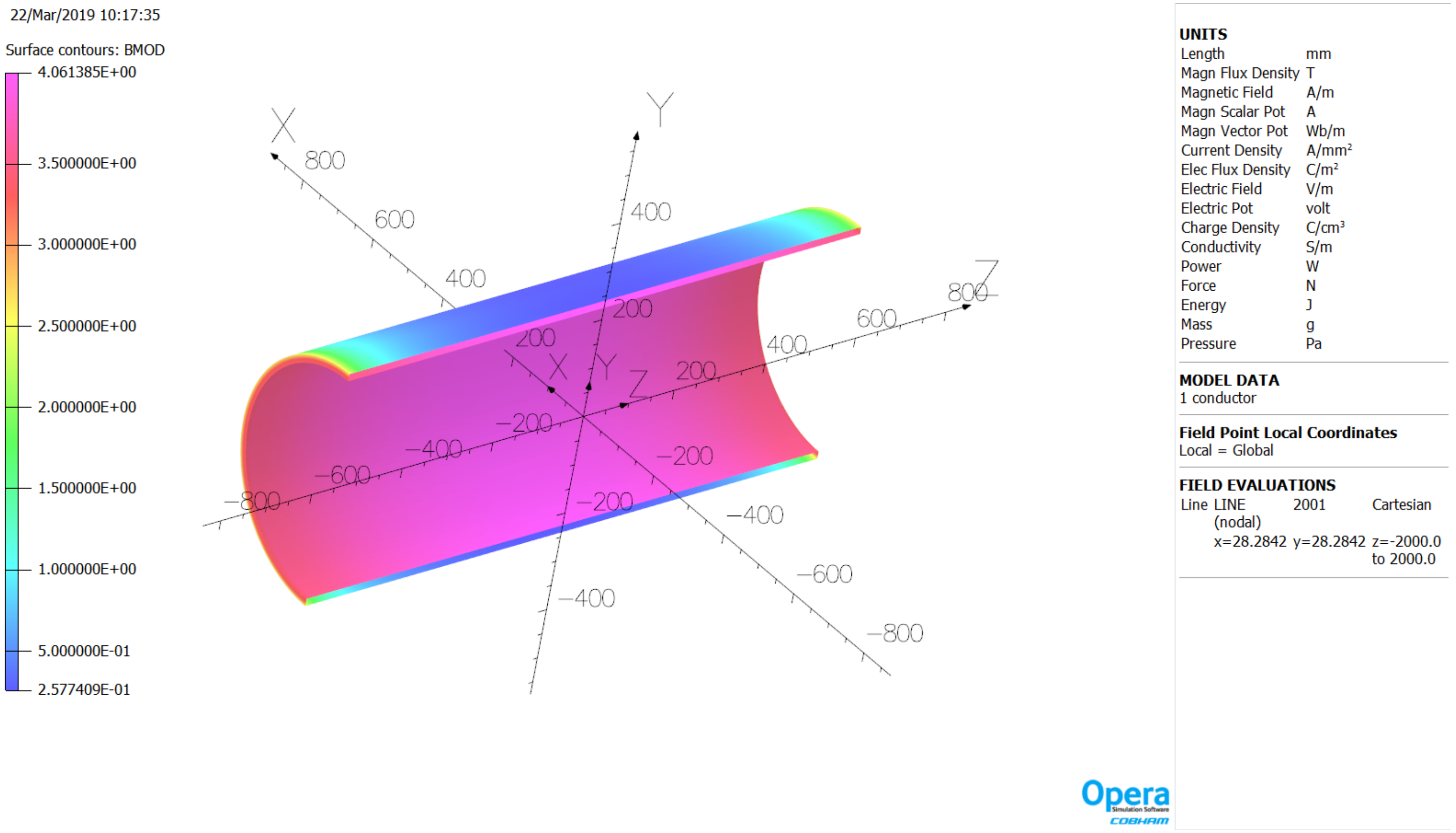
Nested Skew Quadrupole



Independent Skew Quadrupole

Solenoid Magnet

- There are four solenoid magnets in the IR to counteract the effects of the detector solenoid
- The downstream ion beam solenoid has a larger bore (19.8 cm radius), coil inner radius (22.5 cm) and it is a 4 T solenoid.
- The small bore solenoids have a central field of 2T and 4T the coil inner radius is 6.0 cm
- The solenoid magnets will use standard MRI NbTi rectangular conductor



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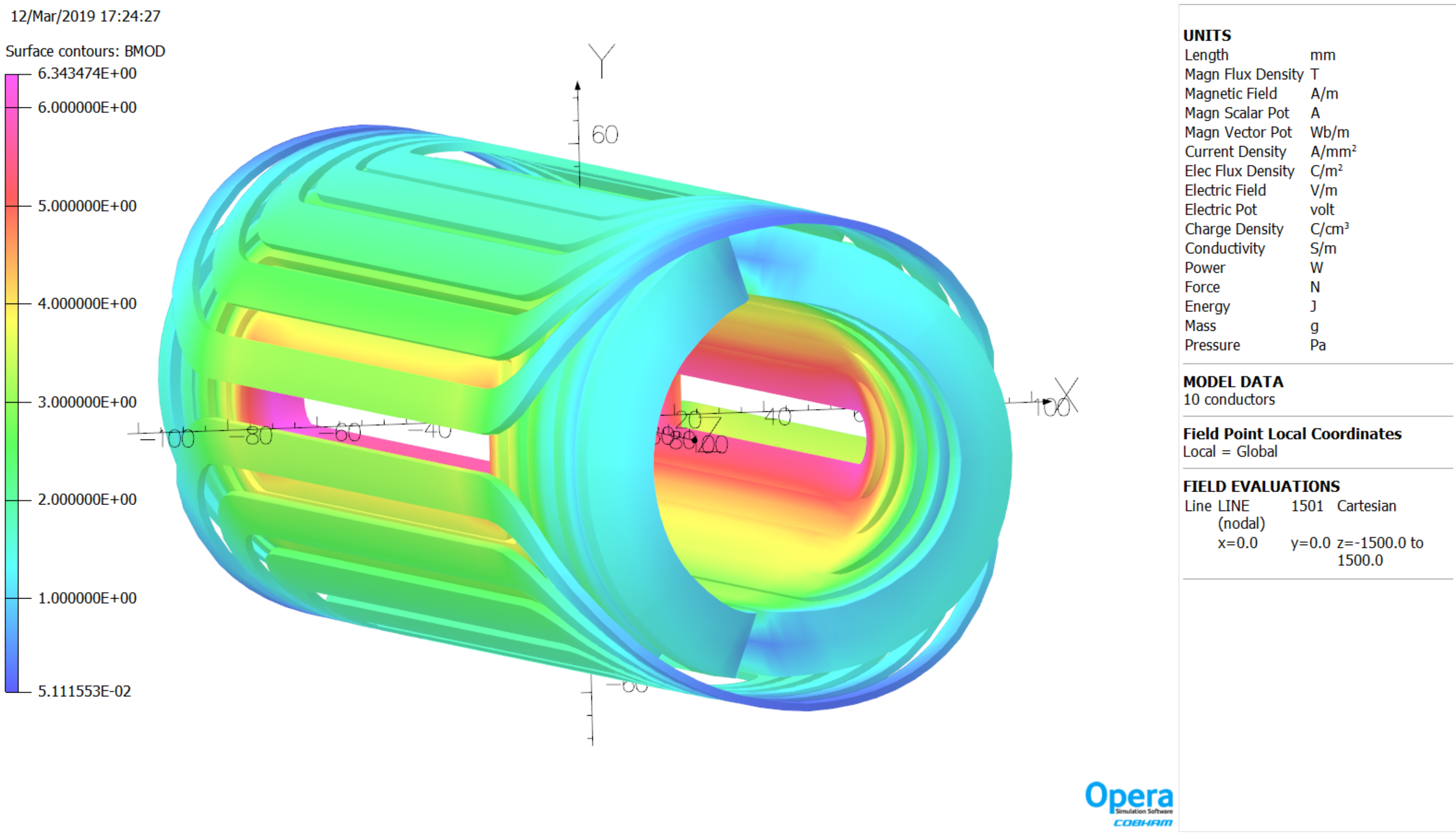
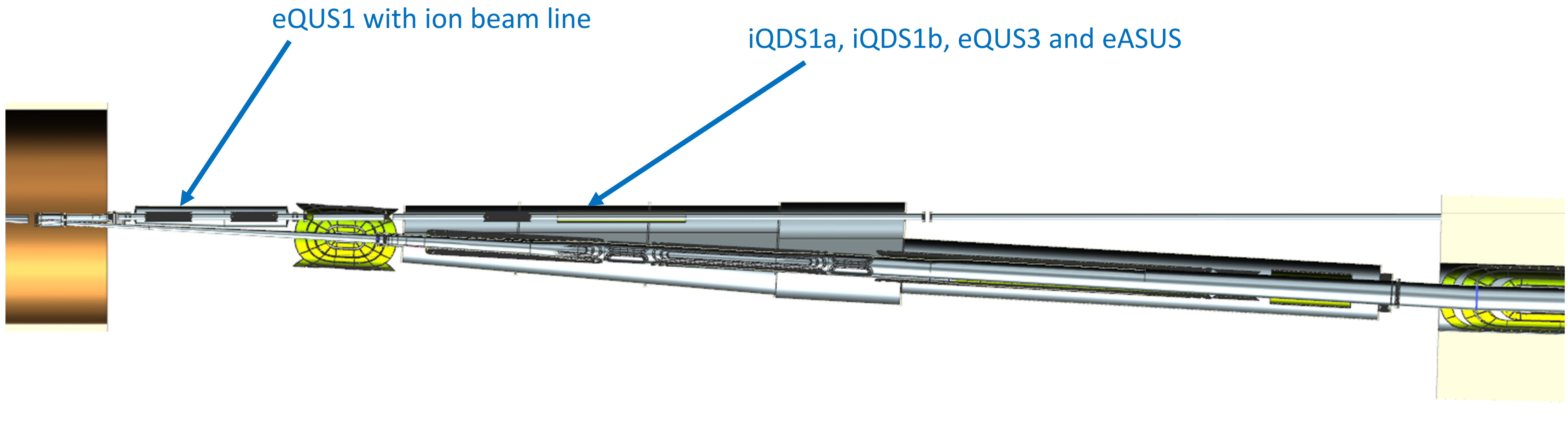
Kicker Magnet

- The kicker magnets, iCUS1 and iCUS2, have slightly different field requirements
- Their field directions are reversed
- The designs are close enough that a single design is utilized

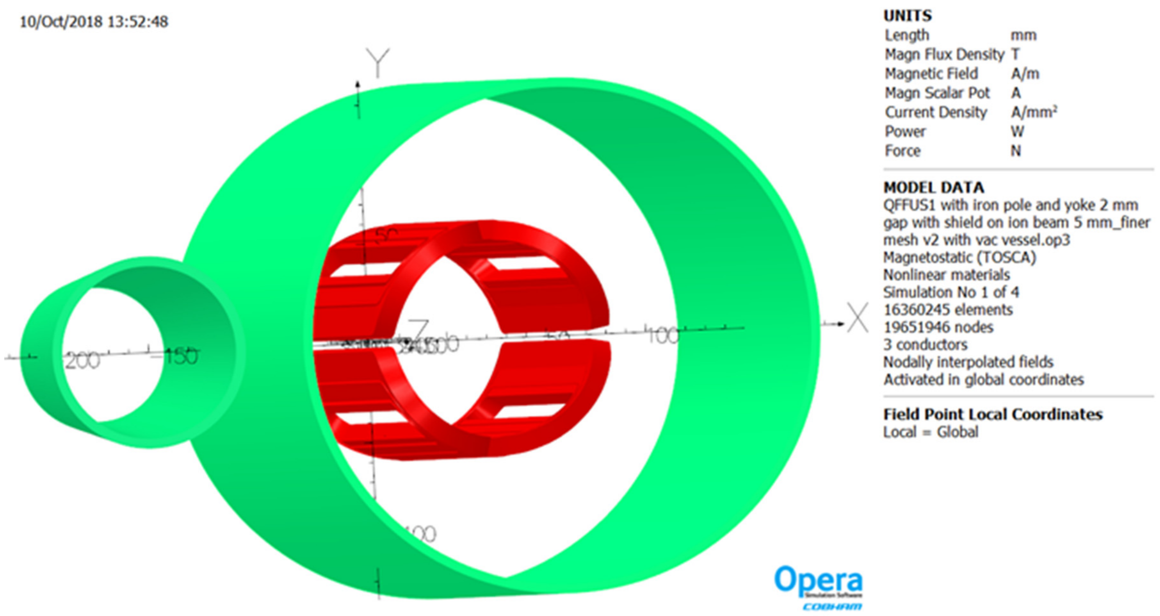


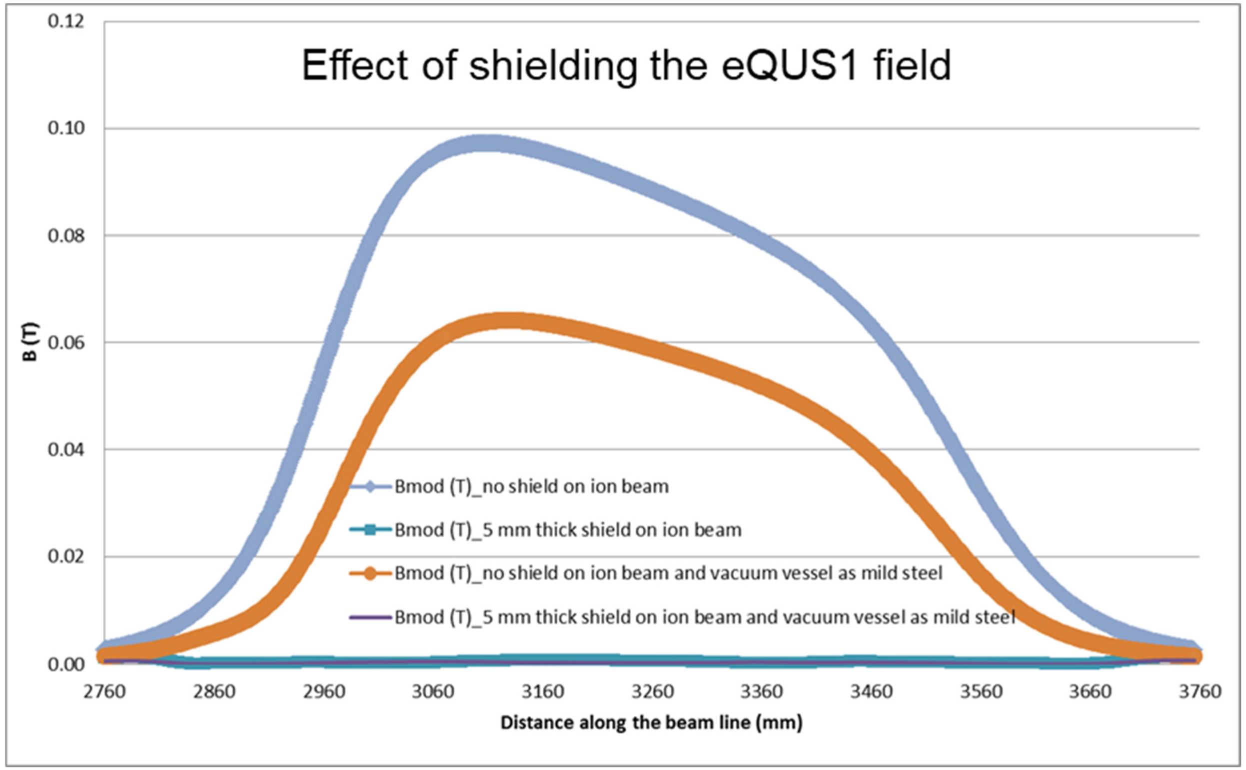
Magnet-Magnet Interaction

- There are more than 24 different magnets in the IR, including nested skew quads, detector solenoid, and detector dipoles
- Due to their proximity to the IP and small crossing angle, these magnets are very close to each other
- The most critical interactions have been studied:

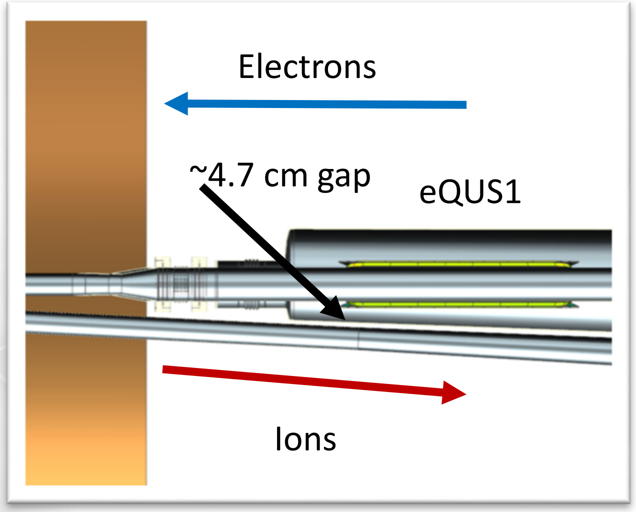


Magnet-Magnet Interaction (eQUS1 with ion beam line)

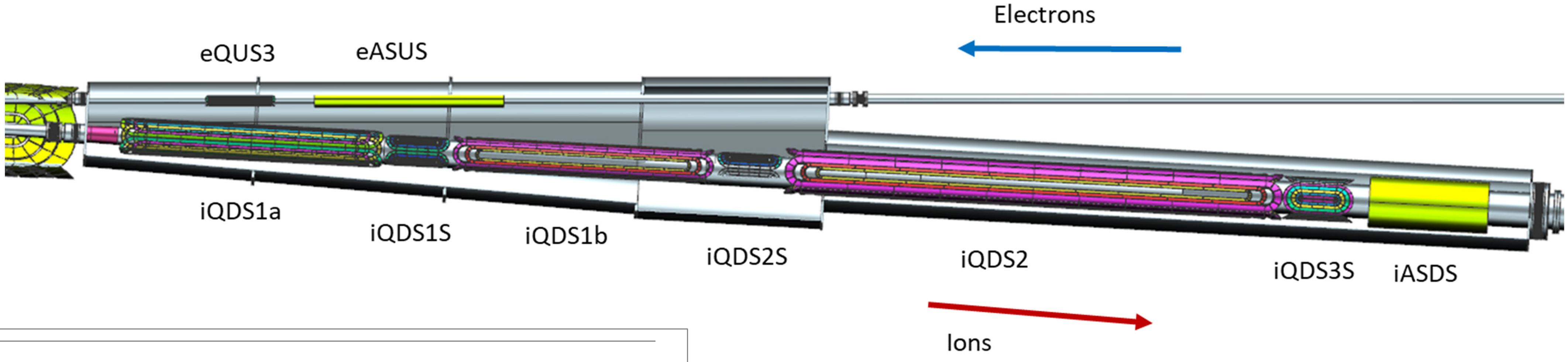


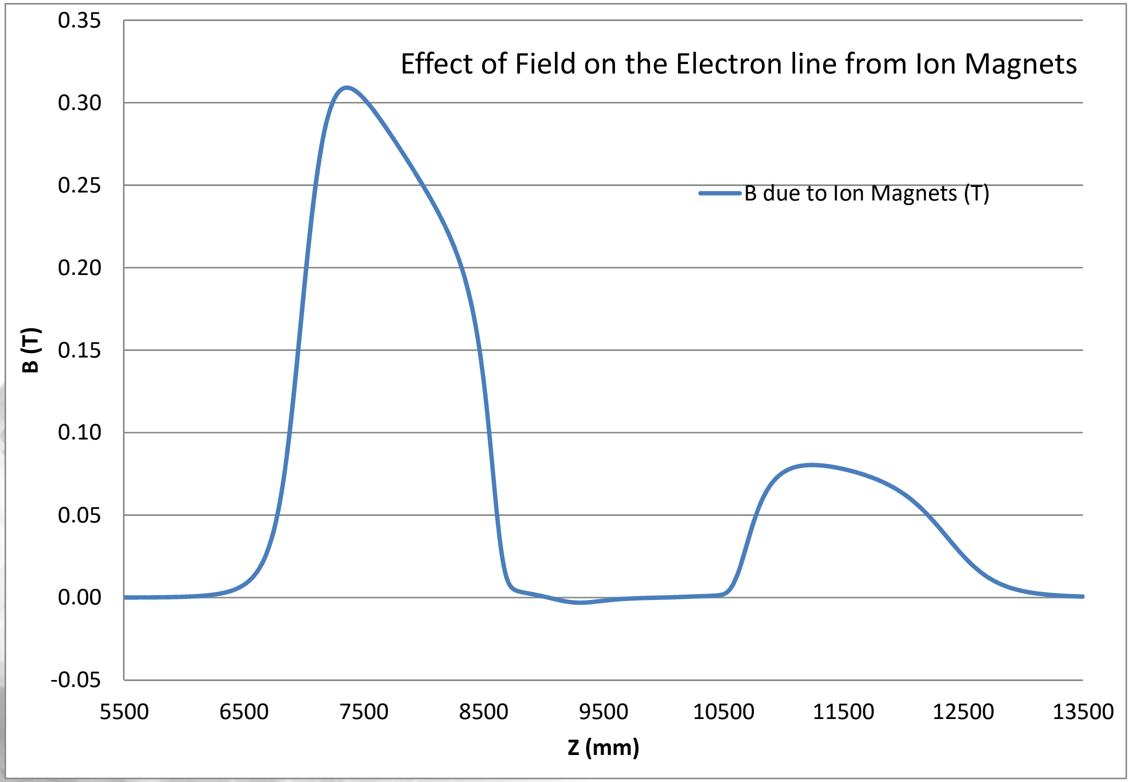


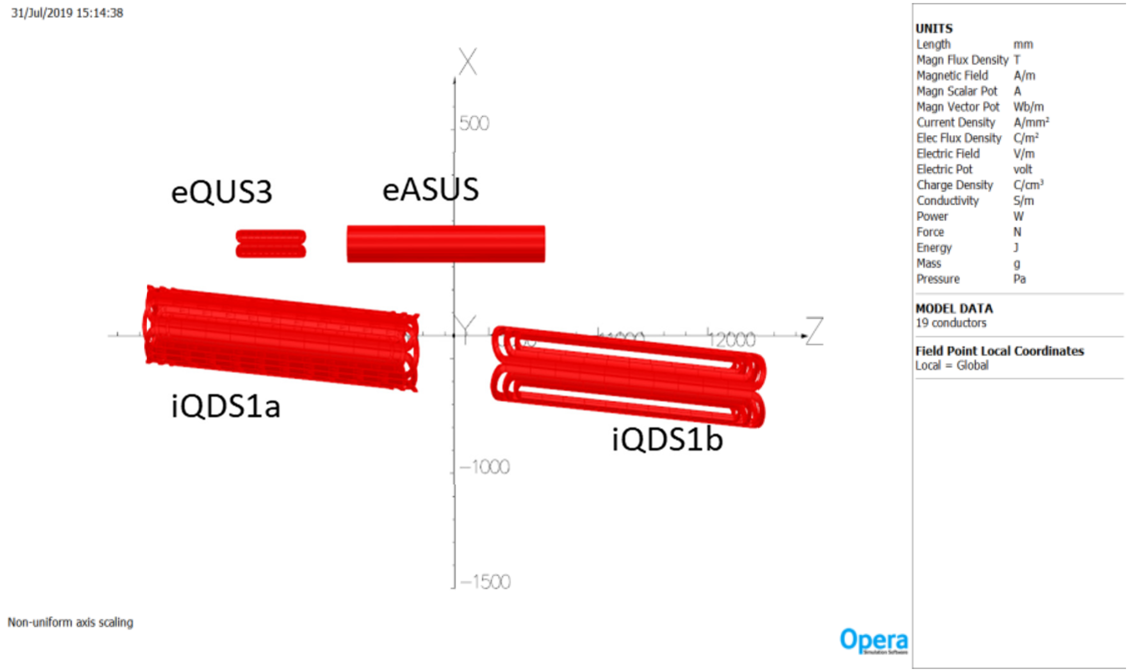
The Coils, Shield around the Ion Beam and Vacuum Vessel around the eQUS1 coils



Magnet-Magnet Interaction (iQDS1a, iQDS1b, eQUS3 and eASUS)







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Summary

- Preliminary designs for all the IR magnets are complete.
- The peak fields in the coils are less than 7 T.
- Acceptable shielding solutions exist for magnet-magnet and magnet-adjacent beamline interactions.
- Preliminary optimizations for the coil peak field have been completed. Additional optimizations are required for detailed coil design, coil peak field, and multipoles.
- The interaction between the detector magnets (main detector solenoid, and two detector dipoles) and the transport magnets remain to be studied.
- The conceptual cryostat design for all IR magnets is in progress.

NAPAC 2019 Reference Presentations and Posters

- MOOHC2 - Electron-Ion Accelerator Designs
- TUZBA2 - EIC Machine Detector Interface
- MOPLO13 - Field Quality Analysis of Interaction Region Quadrupoles for JLEIC
- TUPLO15 - Multipole Effects on Dynamic Aperture in JLEIC Ion Collider Ring

Thank you for your attention.

Questions?